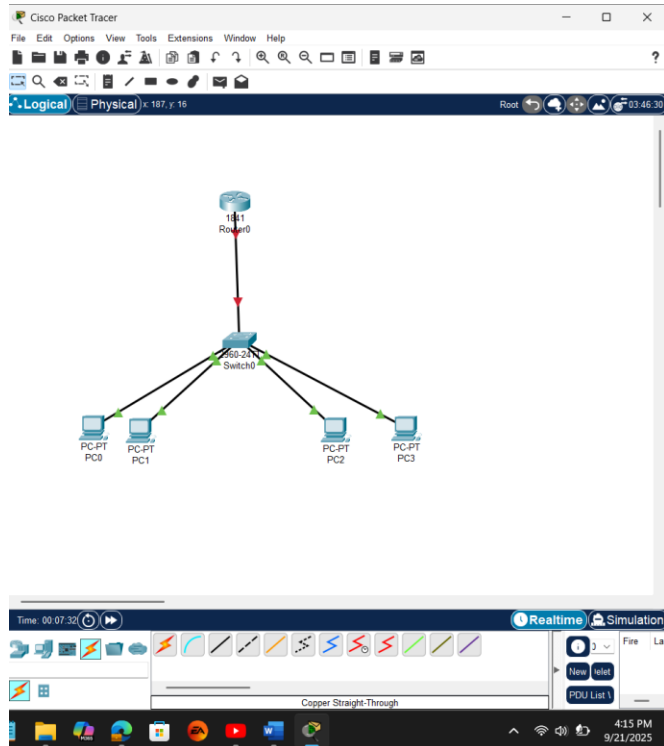


NET622.FA3.WEEK8



Router0

Physical Config CLI Attributes

FastEthernet0/0

Port Status: ☒ On
Bandwidth: 100 Mbps ☒ 10 Mbps ☒ Auto
Duplex: ☒ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address: 0060.3E06.C701

IP Configuration
IPv4 Address: 192.168.1.1
Subnet Mask: 255.255.255.0

Tx Ring Limit: 10

Equivalent IOS Commands

```
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

☐ Top

PC0

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC0

InterfacesFastEthernet0

Gateway/DNS IPv4

DHCP

Static

Default Gateway192.168.1.1

DNS Server

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

Top

PC0

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

Bandwidth

Duplex

MAC Address00E0.8F14.0D4E

IP Configuration

DHCP

Static

IPv4 Address192.168.1.2

Subnet Mask255.255.255.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address: FE80::2E0:8FFF:FE14:D4E

Top

PC1

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC1

Interfaces FastEthernet0

Gateway/DNS IPv4

DHCP

Static

Default Gateway 192.168.1.1

DNS Server

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

☐ Top

PC1

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth

100 Mbps10 Mbps

☒ Auto

Duplex

Half DuplexFull Duplex

☒ Auto

MAC Address 0001.64BE.4378

IP Configuration

DHCP

Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address: FE80: 201:64FF:FEBE:4378

☐ Top

PC2

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC2

Interfaces FastEthernet0

Gateway/DNS IPv4

DHCP

Static

Default Gateway 192.168.20.1

DNS Server

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

Top

PC2

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

Bandwidth

Duplex

MAC Address

0005.5E49.1C09

IP Configuration

DHCP

Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address: FE80::205:5EFF:FE49:1C09

Top

PC3

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC3

Interfaces FastEthernet0

Gateway/DNS IPv4

DHCP

Static

Default Gateway 192.168.20.1

DNS Server

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

☐ Top

PC3

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

Bandwidth

Duplex

MAC Address 00E0.A3D3.23DA

IP Configuration

DHCP

Static

IPv4 Address 192.168.20.3

Subnet Mask 255.255.255.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address: FE80::2E0:A3FF:FE03:23DA

☐ Top

Switch0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

VLAN Configuration

VLAN Number20

VLAN Namelab1

AddRemove

VLAN No

VLAN Name

1default

20lab1

1002fdi-default

1003token-ring-default

1004fddinet-default

1005tmet-default

Equivalent IOS Commands

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)# name lab1
Switch(config-vlan)#
```

Top

Switch0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

FastEthernet0/4

Port Status

Bandwidth

Duplex

Trunk

Trunk

Access

VLAN

1

10

Equivalent IOS Commands

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/4
Switch(config-if)#
%SYS-5-CONFIG_I: Configured from console by console
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
```

Top

Switch0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

Port Status

Bandwidth

Duplex

Access

VLAN

10

Tx Ring Limit

20

1 default

20 lab1

1002 fddi-default

On

Auto

100 Mbps

10 Mbps

Half Duplex

Full Duplex

Auto

Equivalent IOS Commands

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#exit
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
```

Top

Switch0

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/16

FastEthernet0/17

Port Status

Bandwidth

Duplex

Access

VLAN

10

Tx Ring Limit

20

1 default

20 lab1

1002 fddi-default

On

Auto

100 Mbps

10 Mbps

Half Duplex

Full Duplex

Auto

Equivalent IOS Commands

```
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/3
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/5
Switch(config-if)#
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
```

Top

Router0

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database**
- INTERFACE

VLAN Configuration

VLAN Number: 20

VLAN Name: lab1

VLAN No	VLAN Name
1	default
20	lab1
1002	fdi-default
1003	token-ring-default
1004	fdinet-default
1005	trnet-default

Equivalent IOS Commands

```
Router>enable
Router#
Router#vlan database
% Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.

Router(vlan)#vlan 20 name lab1
VLAN 20 modified:
  Name: lab1
Router(vlan)#
```

☐ Top

Router0

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

```
Router>enable
Router#
Router#vlan database
% Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.

Router(vlan)#vlan 20 name lab1
VLAN 20 modified:
  Name: lab1
Router(vlan)#
Router(vlan)#exit
APPLY completed.
Exiting....
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface 0/0
^
% Invalid input detected at '^' marker.

Router(config)#int f 0/0
Router(config-if)#exit
Router(config)#int f 0/0.1
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.1, changed state to up

Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

☐ Top

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x 1563, y 207

```
graph TD
    Router[Router 192.168.1.1] --- Switch[Switch 192.168.20.24]
    Switch --- PC1[PC-PT PC0 192.168.1.2]
    Switch --- PC2[PC-PT PC1 192.168.1.3]
    Switch --- PC3[PC-PT PC2 192.168.20.2]
    Switch --- PC4[PC-PT PC3 192.168.20.3]
```

Time: 00:40:16

Scenario 1

Toggle PDU List Window

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=1ms TTL=127
Reply from 192.168.20.3: bytes=32 time=1ms TTL=127
Reply from 192.168.20.3: bytes=32 time=1ms TTL=127
Reply from 192.168.20.3: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

28°C Sunny 4:48 PM 9/21/2025